

OS SKILL 8

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_VARS 20

typedef struct {
    char name[50];
    char value[100];
} Variable;

Variable vars[MAX_VARS];
int var_count = 0;

/* Store or update variable */
void set_variable(const char *name, const char *value) {
    for (int i = 0; i < var_count; i++) {
        if (strcmp(vars[i].name, name) == 0) {
            strcpy(vars[i].value, value);
            return;
        }
    }

    strcpy(vars[var_count].name, name);
    strcpy(vars[var_count].value, value);
    var_count++;
}

/* Find variable value */
char *get_variable(const char *name) {
    for (int i = 0; i < var_count; i++) {
        if (strcmp(vars[i].name, name) == 0)
            return vars[i].value;
    }

    return NULL;
}
```

```
/* Expand a variable */
void expand_variable(const char *name) {
    char *value = get_variable(name);

    if (value != NULL)
        printf("%s = %s\n", name, value);
    else
        printf("%s is undefined\n", name);
}

/* Built-in: cd */
void builtin_cd(char *path) {
    if (chdir(path) == 0)
        printf("Directory changed successfully\n");
    else
        perror("cd");
}

/* Built-in: pwd */
void builtin_pwd() {
    char cwd[1024];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("Current directory: %s\n", cwd);
}

/* Built-in: echo */
void builtin_echo(char *text) {
    printf("%s\n", text);
}
```

```
/* Dispatch built-in commands */
void dispatch(char *command, char *argument) {

    if (strcmp(command, "pwd") == 0) {
        builtin_pwd();
    }

    else if (strcmp(command, "cd") == 0) {
        builtin_cd(argument);
    }

    else if (strcmp(command, "echo") == 0) {
        builtin_echo(argument);
    }

    else {
        printf("Invalid command: %s\n", command);
    }
}
```

```

int main() {

    printf("=== Variable Reference Detection ===\n");

    set_variable("USER_NAME", "Nishanth");
    set_variable("OS", "Ubuntu");

    printf("Detected variable reference: $USER_NAME\n");
    expand_variable("USER_NAME");

    printf("\n=== Expand Variable Values ===\n");
    expand_variable("OS");

    printf("\n=== Undefined Variable Test ===\n");
    expand_variable("UNKNOWN");

    printf("\n=== Update Variable ===\n");
    set_variable("OS", "Linux");
    expand_variable("OS");

    printf("\n=== Nested Variable Test ===\n");

    set_variable("A", "B");
    set_variable("B", "NestedValue");

    char *first = get_variable("A");

    if (first != NULL) {
        printf("A -> %s\n", first);

        char *second = get_variable(first);

        if (second != NULL)
            printf("Nested expansion: %s\n", second);
    }

    printf("\n=== Built-in Command Dispatch ===\n");

    dispatch("pwd", "");

    dispatch("echo", "Hello from built-in echo");

    printf("\n=== Maintain State Test ===\n");
    dispatch("cd", "/tmp");
    dispatch("pwd", "");
}

```

```

printf("\n=== Built-in Command Dispatch ===\n");

dispatch("pwd", "");

dispatch("echo", "Hello from built-in echo");

printf("\n=== Maintain State Test ===\n");
dispatch("cd", "/tmp");
dispatch("pwd", "");

printf("\n=== Invalid Command Test ===\n");
dispatch("wrongcommand", "");

printf("\n=== Dispatch Logic Test Complete ===\n");

return 0;
}

```

Output:

```

nishanth@LAPTOP-6Q3ON07H: ~/task8
nishanth@LAPTOP-6Q3ON07H:~$ mkdir task8
nishanth@LAPTOP-6Q3ON07H:~$ cd task8
nishanth@LAPTOP-6Q3ON07H:~/task8$ nano task8.c
nishanth@LAPTOP-6Q3ON07H:~/task8$ gcc task8.c -o task8
nishanth@LAPTOP-6Q3ON07H:~/task8$ ./task8
=== Variable Reference Detection ===
Detected variable reference: $USER_NAME
USER_NAME = Nishanth

=== Expand Variable Values ===
OS = Ubuntu

=== Undefined Variable Test ===
UNKNOWN is undefined

=== Update Variable ===
OS = Linux

=== Nested Variable Test ===
A => B
Nested expansion: NestedValue

=== Built-in Command Dispatch ===
Current directory: /home/nishanth/task8
Hello from built-in echo

=== Maintain State Test ===
Directory changed successfully
Current directory: /tmp

=== Invalid Command Test ===
Invalid command: wrongcommand

=== Dispatch Logic Test Complete ===
nishanth@LAPTOP-6Q3ON07H:~/task8$

```